# Vilém Zouhar

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## Education

**ETH Zürich** 2022-Dec 2026 (expected)
PhD in Computer Science in **LRE lab**
Advisors: Mrinmaya Sachan, Menna El-Assady
Thesis: Prudent & Bespoke AI Evaluation

**Saarland University + Groningen University** Sep 2020 - Jun 2022
MSc. in Language Science and Technology (double)
Advisors: Dietrich Klakow, Gosse Bouma
Thesis **Shrinking Knowledge Base Size: Dimension Reduction & Filtering**

**Charles University in Prague** Sep 2017 - Jun 2020
BSc. in Computer Science (computational linguistics minor)
Advisor: Ondřej Bojar
Thesis **Enabling Outbound Machine Translation**

## Research Interests (inter alia)

- Formal methods of AI evaluation and safety — *„Without evaluation, science is limited to blind exploration.“*
- Practical AI evaluation also with humans — *„How to make humans judge quality?“*
- Efficient AI evaluation with applied statistics — *How to compare models robustly yet economically?*

## Work Experience

- **Google Translate**, Mountain View (research intern, evaluation) Jun-Oct 2025
- Microsoft Translate, remote (applied scientist intern, model evaluation) Jun-Oct 2024
- **Amazon Machine Translation**, New York (applied scientist intern, quality estimation) Jun-Oct 2023
- **Spoken Language Systems**, Saarland University (student research assistant) Jan-Dec 2021
- **Institute of Formal and Applied Linguistics**, Charles University (student research assist.) Oct 2019 - Jun 2022
- Previo (software dev intern) Jun-Sep 2018
- BIM Project (software dev intern) Jun-Sep 2017

## Awards and Scholarships

- Cohere Labs Catalyst grant for efficient evaluation 2026
- Google PhD Fellowship in Natural Language Processing 2025
- Two EAMT grants for research on compute-efficient evaluation, and LLM brittleness 2025, 2025
- Language & Communication Technologies full MSc Erasmus Mundus **scholarship** 2020-2022
- Student faculty grant for **learning materials** for Machine Learning Exercises in R 2018

## Teaching

- Human evaluation tutorial EAMT 2026, KONVENS 2026, MT Marathon 2026
- Large Language Models TA **2023**, **2024**, **2025**, **2026** (+lecture)
- Deep Learning TA **2025**
- Interactive Machineimon Su Learning Visualization and Explainability TA **2025**
- Neural Networks Implementation and Applications head TA **2021**
- Statistical Natural Language Processing head TA **2021**

## Selected Publications

Google Scholar Semantic Scholar

**Contrastive ESA: Human Evaluation of Multiple Translations at Once** (WMT 2026)

**Vilém Zouhar**, Roman Grundkiewicz, Sara Rajaee, Parker Riley, Martin Popel, Rachel Bawden, Philipp Koehn, Marine Carpuat, Tom Kocmi

**Pearmut: Human Evaluation of Translation Made Trivial** (EMNLP 2026 Demo)

**Vilém Zouhar**, Tom Kocmi

**How to Select Datapoints for Efficient Human Evaluation of NLG Models?** (TACL 2025)

**Vilém Zouhar**, Peng Cui, Mrinmaya Sachan

**Early-Exit and Instant Confidence Translation Quality Estimation** (EACL 2026)

**Vilém Zouhar**, Maike Züfle, Beni Egressy, Julius Cheng, Jan Niehues

**Searching for Difficult-to-Translate Test Examples at Scale** (In review 2026)

Wenda Xu,[=] **Vilém Zouhar**,[=] Parker Riley, Mara Finkelstein, Markus Freitag, Daniel Deutsch

**Pitfalls and Outlooks in Using COMET** (WMT 2024)

**Vilém Zouhar**,[=] Pinzhen Chen,[=] Tsz Kin Lam, Nikita Moghe, Barry Haddow

**Fine-Tuned Machine Translation Metrics Struggle in Unseen Domains** (ACL 2024)

**Vilém Zouhar**, Shuoyang Ding, Anna Currey, Tatyana Badeka, Jenyuan Wang, Brian Thompson

**RELIC: Investigating Large Language Model Responses using Self-Consistency** (CHI 2024)

Furui Cheng, **Vilém Zouhar**, Simran Arora, Mrinmaya Sachan, Hendrik Strobelt, Mennatallah El-Assady

**Evaluating Optimal Reference Translations** (JNLE 2024)

**Vilém Zouhar**, Věra Kloudová, Martin Popel, Ondřej Bojar

**Tokenization and the Noiseless Channel** (ACL 2023)

**Vilém Zouhar**, Clara Meister, Juan Luis Gastaldi, Li Du, Mrinmaya Sachan, Ryan Cotterell

**Re-visiting Automated Topic Model Evaluation with Large Language Models** (EMNLP 2023)

Dominik Stammbach, **Vilém Zouhar**, Alexander Hoyle, Mrinmaya Sachan, Elliott Ash

**Neural Machine Translation Quality and Post-Editing Performance** (EMNLP 2021)

**Vilém Zouhar**, Ondřej Bojar, Martin Popel, Aleš Tamchyna

**Dynamically Allocating Evaluation Effort for Model Ranking** (EMNLP 2026)

**Vilém Zouhar**, Julia Kreutzer, Alon Lavie, Tom Kocmi, Matt Post, Ondřej Bojar, Mrinmaya Sachan

**Last Translation Benchmark** (in progress 2026)

**Vilém Zouhar** and >200 co-authors

**Estimating Machine Translation Difficulty** (EMNLP 2025)

Lorenzo Proietti,[=] Stefano Perrella,[=] **Vilém Zouhar**,[=] Roberto Navigli, Tom Kocmi

**Generating Difficult-to-Translate Texts** (MME at EACL 2026)

**Vilém Zouhar**, Wenda Xu, Parker Riley, Juraj Juraska, Mara Finkelstein, Markus Freitag, Dan Deutsch

**AI-Assisted Human Evaluation of Machine Translation** (NAACL 2025)

**Vilém Zouhar**,[=] Tom Kocmi,[=] Mrinmaya Sachan

**Error Span Annotation: A Balanced Approach for Human Evaluation of Machine Translation** (WMT 2024)

Tom Kocmi,[=] **Vilém Zouhar**,[=] Eleftherios Avramidis, Roman Grundkiewicz, Marzena Karpinska, Maja Popović, Mrinmaya Sachan, Mariya Shmatova

**Navigating the Metrics Maze: Reconciling Score Magnitudes and Accuracies** (ACL 2024)

Tom Kocmi, **Vilém Zouhar**, Christian Federmann, Matt Post

**Distributional Properties of Subword Regularization** (EMNLP 2024)

Marco Cognetta, **Vilém Zouhar**, Naoaki Okazaki

**A Diachronic Perspective on User Trust in AI under Uncertainty** (EMNLP 2023)

Shehzaad Dhuliawala,[=] **Vilém Zouhar**,[=] Mennatallah El-Assady, Mrinmaya Sachan

**A Formal Perspective on Byte-Pair Encoding** (ACL 2023)

**Vilém Zouhar**, Clara Meister, Juan Luis Gastaldi, Li Du, Tim Vieira, Mrinmaya Sachan, Ryan Cotterell

**Poor Man's Quality Estimation: Predicting Ref.-Based MT Metrics Without Reference** (EACL 2023)

**Vilém Zouhar**, Shehzaad Dhuliawala, Wangchunshu Zhou, Nico Daheim, Tom Kocmi, Yuchen Eleanor Jiang, Mrinmaya Sachan

**Providing Backtranslation Improves Users Confidence in MT, Not Quality** (NAACL 2021)

**V. Zouhar**, M. Novák, M. Žilinec, O. Bojar, M. Obregón, R. L. Hill, F. Blain, M. Fomicheva, L. Specia, L. Yankovskaya

## Other Projects and Shared Tasks

GitHub

**Apertus 1.5: From Open Text Models to Open Multimodal Reasoners** (preprint 2026)

**Why We Need Speech to Evaluate Speech Translation** (EMNLP 2026)

**There is No Theoretical Curse of Multilinguality For Embedding Space Structure** (in review 2026)

**Augmenting Text to Increase Translation Difficulty** (EAMT 2026)

**When AI Benchmarks Plateau: A Systematic Study of Benchmark Saturation** (ICML 2026)

**Findings of the WMT26 Shared Task on Automated Translation Quality Evaluation Systems: Compact Open Models are Competitive Quality Evaluators** (WMT 2026)

Alon Lavie, **V. Zouhar** and others

**Findings of the WMT26 Terminology Translation Task: The Hard Part is Finding the Terms, Not Using Them** (WMT 2026)

Adrian Charkiewicz, **V. Zouhar** and others

**The Role of Disfluencies in Speech Translation** (in review 2026)

**Hearing to Translate: The Effectiveness of Speech Modality Integration into LLMs** (TACL 2026)

**Findings of the WMT25 shared task on automated translation evaluation systems: Linguistic diversity is challenging and references still help** (WMT 2025)

Alon Lavie, **V. Zouhar** and others

**COMET-poly: Machine Translation Metric Grounded in Other Candidates** (WMT 2025)

**Co-DETECT: Collaborative Discovery of Edge Cases in Text Classification** (EMNLP 2025 Demo)

**Large Language Models as Span Annotators** (MME at EACL 2026)

**A Bayesian Optimization Approach to Machine Translation Reranking** (NAACL 2025)

**WMT 2023 Shared Task on Machine Translation with Terminologies** (EMNLP 2023)

Kirill Semenov, **Vilém Zouhar**, Tom Kocmi, Dongdong Zhang, Wangchunshu Zhou, Yuchen Eleanor Jiang

**Speech Translation and Metrics in 2026: Findings of the IWSLT Campaign** (IWSLT 2026)

David Ifeoluwa Adelani, **V. Zouhar** and others

**Are Large Language Models for Education Reliable for All Languages?** (BEA 2025)

**PWESuite: Phonetic Word Embeddings and Tasks They Facilitate** (LREC-COLING 2024)

**How to Engage Your Readers? Generating Guiding Questions to Promote Active Reading** (ACL 2024)

**Enhancing Textbooks with Visuals from the Web for Improved Learning** (EMNLP 2023)

**Findings of the WMT26 General Machine Translation Shared Task: Contrastive Dynamic Human Evaluation at Scale** (WMT 2026)

Tom Kocmi, **V. Zouhar** and others

**Findings of the WMT26 Multilingual Instruction Shared Task: Small Models Are Not Yet Polyglots** (WMT 2026)

Patrícia Schmidtová, **V. Zouhar** and others

**Findings of the WMT25 general machine translation shared task: Time to stop evaluating on easy test sets** (WMT 2025)

Tom Kocmi, **V. Zouhar** and others

**Can Large Language Models Capture Human Annotator Disagreements?** (EACL 2026)

**Findings of the WMT25 Multilingual Instruction Shared Task: Persistent Hurdles in Reasoning, Generation, and Evaluation** (WMT 2025)

Tom Kocmi, **V. Zouhar** and others

**Findings of the WMT25 Terminology Translation Task: Terminology is Useful Especially for Good MTs** (WMT 2025)

Kirill Semenov, Xu Huang, **Vilém Zouhar**, Nathaniel Berger, Dawei Zhu, Arturo Oncevay, Pinzhen Chen

**Deconstructing Self-Bias in LLM-generated Translation Benchmarks** (EMNLP 2026)

**Unsupervised Word-level Quality Estimation for Machine Translation Through the Lens of Annotators (Dis)agreement** (EMNLP 2025)

**QE4PE: Word-level Quality Estimation for Human Post-Editing** (TACL 2025)

**WMT24 General Machine Translation Shared Task: The LLM Era is Here but MT is Not Solved Yet** (WMT 2024)

Tom Kocmi, **V. Zouhar** and others

**Hurdles of Automatic Metric for Speech Translation Evaluation** (IWSLT 2026)

**Findings of the IWSLT 2025 Evaluation Campaign** (IWSLT 2025)

Victor Agostinelli, **V. Zouhar** and others

**Understanding Large Language Model Behaviors Through Interactive Counterfactual Generation and Analysis** (IEEEvis 2025)

**Two Counterexamples to Tokenization and the Noiseless Channel** (LREC-COLING 2024)

**AutoTutor meets Large Language Models: A Language Model Tutor with Rich Pedagogy and Guardrails** (Learning@Scale 2024)

## Invited talks

- Theory and Practice of of Efficient Evaluation at Google Gboard (2026), SICSA (2026), Sapienza (2026)
- Selecting Examples to Human-Evaluate NLG Models at Charles University (2025), TU Darmstadt (2025), UMD Multilingual Seminar (2026)
- Prudent NLG Evaluation at KIT (2025), Cardiff NLP Seminars (**2025**), Google Translate (2024)
- Token(s) of Appreciation for BPE at MT Marathon (**2024**)
- ESA and ESAAI at Microsoft Translate (2024)
- How we solved tokenization but got it wrong at ZurichNLP Meetup #9 (**2024**)
- Quality and Quantity of Machine Translation References for Automated Metrics at IST/Unbabel seminar (**2024**)
- Poor Man's Quality Estimation at IST/Unbabel seminar (**2023**)

## Service

- **General Machine Translation** Shared Task at WMT (co-organizer) 2024, 2025, 2026
- **Evaluation Metrics** Shared Task at WMT (co-organizer) 2025, 2026
- **Multilingual Instruction** Shared Task at WMT (co-organizer) 2025, 2026
- **Multilingual Multicultural Evaluation Workshop** at EACL (organizer) 2026
- **SIGTYP Workshop: Computational Linguistic Typology and Multilingual NLP** at EACL (co-organizer) 2026
- **ACL Mentorship Panel**: Navigating Mental Health in Fast-Paced AI Research (panelist) 2026
- **WMT Terminology** Shared Task (organizer) 2023, 2025
- Evaluation committee member for granting university accreditations in Czechia 2020-2025
- **CSRR**: Workshop on Commonsense Representation and Reasoning (organizer) ACL 2022
- Area chair GEM 2026, EMNLP Demo 2026, ARR 2026
- Reviewer: {EACL, MME, ACL, ARR, SwissText, GEM, EMNLP Demo, ARR}2026 {EAMT, CHI} 2027 {NAACL★, ACL, EMNLP, WMT, ARR}2025 {LREC-COLING, EACL★, ARR, Repl4NLP, EMNLP, WMT}2024 {ARR, WMT, EMNLP, AACL-IJCNLP, ACL}2023 {SVRHM, CoNLL, AACL-IJCNLP}2022

## Technical Knowledge

**GitHub**

- **Programming:** Python, JS/TS, Rust
- **Toolkits:** LM eval harness, vllm, PyTorch, Scikit, Numpy, HF, Fairseq, Marian
- **Misc:** Linux (long-term user, cluster), passionate about visualization, writing and typesetting (Typst, LaTeX)

## Languages

- Czech: C2
- English: C2
- German: B2

+ linguistic curiosity

## Students

Pleasure to (co-)supervise student projects/theses

- Rosamund Ang: Sample-Efficient Aligning of Automatic Metrics ETH 2026
- Šimon Sukup, William Kalikman, Michal Tešnar: Augmenting Texts to Increase Translation Difficulty ETH 2026
- Victor Zarzu: Speech Translation Quality Estimation ETH 2026
- Chenfei Xiong: Collaborative Discovery of Edge Cases in Text Classification ETH 2025
- Oymak Gül, Hatipoglu Öykü, Garries Urbina Oscar, Victor Zarzu Interactive Visualization of Chain-of-Thought Reasoning in Large Language Models ETH 2025
- Yijie Tong, Haokun He (Self Quality Estimation for Machine Translation and NLP) UZH 2024
- Abhinav Kumar (Data Augmentation for Text Complexity Prediction) UZH 2024
- Vincent Dörig (IRead: AI-Enhanced Textbook Reading) ETH 2024
- David Gu (Manifestations of Image Complexity) ETH 2023

## Extra-Curricular

- Academic senate at Charles University Faculty of Mathematics and Physics 2018-2020
- Several **games** programmed and presented in limited time, mostly Ludum Dare 2015-present
- Organization of **Kasiopea**, an annual coding competition for talented high school students 2017-2019
- Amateur electric guitar  2021-present